

LC Method Explorer

A local LC method-ranking prototype

Which supported method places a molecule inside a target retention window?

Explore

Input

☒ Held-out example
☐ New SMILES

Compound

Pirimicarb · 52

Model

XGBoost

XGBoost was selected in development.
Other models are comparisons.

Target window (min)

Lower: 5.00 Upper: 10.00

☐ Reveal measurements

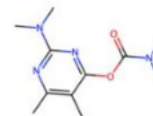
MCMRT V3 · 30 measured methods

LC Method Explorer

Predict retention. Compare measured methods.

Recommendations Evidence

Pirimicarb



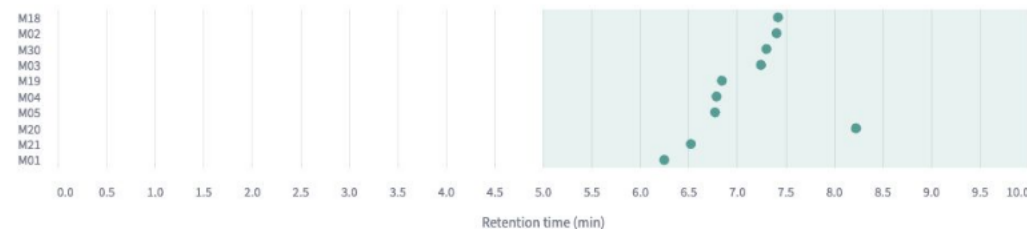
Held-out example · 30/30 methods measured

This compound group was excluded from model fitting. Reveal outcomes to inspect the recommendation.

Formula: C₁₁H₁₈N₄O₂ · Molecular weight: 238.3 · TPSA: 58.6 Å²

Ranked predictions

Top 10 of 30 methods. Shaded band: target window. Points: model predictions.



MCMRT V3 covers the required condition axes

10,073

measured retention times

343

source compounds

30

recorded methods

Condition	Observed coverage
Column products	6 within the RP C18 family
Mobile-phase pairs	6 A/B pairs
Time / %B schedules	15 distinct schedules
Column temperature	30, 40, 45 °C
Total run duration	10–100 minutes

330 of 343

compounds have
measurements
under every method.

217 missing pairs stay
unobserved.

Compound groups define the evaluation

Keep each compound group together across all methods.

Partition	Compounds / groups	Observations	Purpose
Development	273 / 272	8,026	Five grouped folds
Final holdout	70 / 69	2,047	One final evaluation

Inputs available before measurement

21 molecular descriptors plus column, solvent, temperature and gradient features.

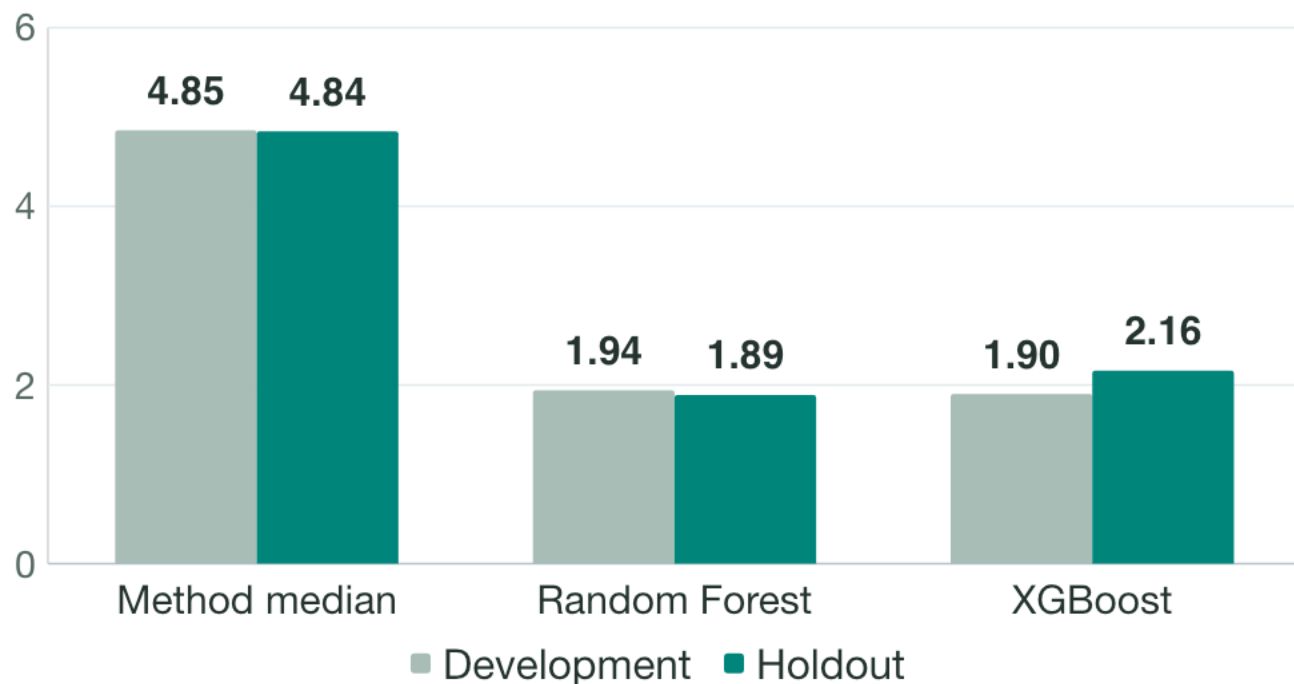
Selection stays inside development

Three configurations per tree family. Select by MAE, freeze the choice, then evaluate the holdout.

Target: unseen compound groups under familiar methods. Grouping is not scaffold separation.

Lower timing error, modest ranking gain

Retention-time MAE (minutes)



XGBoost remains the default

It won development validation. Random Forest did better on the final holdout.

172 vs 169

target-window hits in 268 cases
XGBoost versus a fixed rule

Three additional hits. Statistical significance is not established.

Ranking: 67 complete-panel compounds × 4 prespecified windows. XGBoost 64.2%, fixed rule 63.1%.

Limits and next experiments

Observed failure

14 negative XGBoost predictions

The app flags raw values and retains them in the reported evaluation.

A future experiment can test positive outputs and calibrated uncertainty.

Evidence still needed

New methods and laboratories

Use independent data to test transfer.

Mixture separation quality

Measure resolution, peak shape and sensitivity in the laboratory.

No claim of laboratory deployment or time savings. The evaluated holdout cannot guide further tuning.